JunGyu Lee

Ph.D. Student, Artificial Intelligence
Graduate School of Artificial Intelligence, Yonsei University

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RESEARCH INTERESTS

Crowd Generation, Social Robot Navigation, World Models

EDUCATION

- **Ph.D. in Engineering, Artificial Intelligence**
  Yonsei University, Seoul, Korea
  Advisor: Prof. Hae-Gon Jeon · Visual AI Lab
- **M.S. in AI Robotics**
  University of Science and Technology (UST), Korea Institute of Science and Technology (KIST) School
  Advisor: Prof. Gi Pyo Nam · Visual Intelligence Group (VIG)
- **B.S. in Big Data Analytics**
  Kookmin University, Seoul, Korea

EXPERIENCE

- **AI Researcher**
  Seoul National University Hospital (SNUH)
  Jun 2025 – Nov 2025
- **AI Researcher**
  Korea Institute of Science and Technology (KIST)
  Jan 2023 – Jun 2025
- **Research Internship**
  Asan Medical Center (AMC)
  Aug 2021 – Nov 2021

PUBLICATIONS

* denotes equal contribution

Under Review

JunGyu Lee, Jisu Shin, Seunghyun Shin, Hae-Gon Jeon
Controllable Crowd Generation through World-Model Planning
arXiv preprint arXiv:2610.09438, 2026 (Under review)
Project · arXiv · Code

JunGyu Lee, Inhwan Bae, Hae-Gon Jeon
Revisiting Numerical Forecasting Models for Language-Based Trajectory Prediction
arXiv preprint arXiv:2610.07954, 2026 (Under review)
Project · arXiv · Code

Boa Jang*, **JunGyu Lee***, Gwanho Lee, Jinwook Choi, Young-Gon Kim
Skeleton-Guided Progressive Test-Time Adaptation for Thin Curvilinear Structures
arXiv preprint arXiv:2610.11104, 2026 (Under review)
Project · arXiv · Code

Sang-Hun Han, Changyeon Won, **JunGyu Lee**, Min-Gyu Park, Hae-Gon Jeon
IORA: Irregular Object Immobilization under Restricted Actuation
arXiv preprint arXiv:2610.30969, 2026 (Under review)
Project · arXiv

2026

Jisu Shin, Junoh Lee, **JunGyu Lee**, Inhwan Bae, Dohyeon Lee, Hokyun Im, Youngwoon Lee, Hae-Gon Jeon
ComPose: When to Trust Hands for Object Pose Tracking

NeurIPS 2026 — Conference on Neural Information Processing Systems (NeurIPS)
Project · arXiv

Dong Yeong Kim*, **JunGyu Lee***, Jaewon Choi, June Young Seo, Myeongseop Kim, Jinwook Choi, Taek Min Kim, Young-Gon Kim
Distilling Temporal Coherence into 2D Networks for TRUS Prostate Video Segmentation
MICCAI 2026 — Proceedings of the International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)
Project · Paper · arXiv · Code

Dong Yeong Kim, Jaewon Choi, Youmin Shin, **JunGyu Lee**, Myeongseop Kim, Jinwook Choi, Joo Whan Kim, Young-Gon Kim
PSCT-Net: Geometry-Aware Pediatric Skull CT Reconstruction via Differentiable Back-Projection and Attention-Guided Refinement
MICCAIW 2026 — Proceedings of the International Conference on Medical Image Computing and Computer-Assisted Intervention Workshop (MICCAIW)
Project · Paper · arXiv · Code

2025

JunGyu Lee, Kunyoung Lee, Haesol Park, Ig-Jae Kim, Gi Pyo Nam
V-NAW: Video-based Noise-aware Adaptive Weighting for Facial Expression Recognition
CVPRW 2025 — Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshop (CVPRW)
Project · Paper · arXiv · Code

JunGyu Lee*, Yeji Choi*, Haksub Kim, Ig-Jae Kim, Gi Pyo Nam
Navigating Label Ambiguity for Facial Expression Recognition in the Wild
AAAI 2025 — Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)
Project · Paper · arXiv · Code

2024

JunGyu Lee, Sanghyun Alex Lee, Gi Pyo Nam
Study on Facial Composite Feature Analysis for Determining Subject Anxiety Levels on Low-Power Computing Modules
IEIE 2024 — Summer Annual Conference of IEIE
Project · Paper · Code

HONORS & AWARDS

- **Top 5 Placement, 8th ABAW Competition @ CVPR Workshop**
  2025
- **Best Paper Award, KIST Convergence Conference**
  Korea Institute of Science and Technology (KIST)
  2024
- **Outstanding Technology Award, AI-Robotics Research Division**
  Korea Institute of Science and Technology (KIST)
  2024

SELECTED PROJECTS

- **Expression Recognition App**
  PyQt-based facial expression recognition from images, videos, or real-time webcam
- **Realtime FER / VA Estimation / Eye Blink Detection / Face Detector**
  Real-time facial analysis pipelines for expression, affect, and behavioral monitoring

MEDIA

- **AI, Reading Inner Feelings**
  Chosun Ilbo · Jul 2023 · Article
- **On-site Verification: Drugs, from Inflow to Distribution — Can They Be Stopped?**
  MBC Newsdesk · May 2023 · Video
- **Customs' Cutting-Edge Detection Technology on the Front Line of the 'War on Drugs'**
  JTBC News (D:Issue) · Apr 2023 · Video